

Kafka Broker Not Available SOP

Objective

Restore Kafka broker availability and message streaming when producers or consumers cannot connect to the Kafka cluster.

Scope

Applicable to Kafka-based applications, streaming platforms, backend services, data engineering pipelines, DevOps teams, and platform teams supporting production or synthetic incident workflows.

Pre-requisites

1. Access to Kafka cluster, namespace or broker host.
2. Access to monitoring dashboards such as Grafana, Prometheus, or cloud monitoring.
3. Kafka CLI tools: kafka-topics, kafka-consumer-groups, kafka-console-producer, and kafka-console-consumer.
4. Permission to view application logs, broker logs, and deployment status.
5. Escalation contact for DevOps, platform, and application owner teams.

Incident Metadata

Field	Value
Ticket ID	T011
Issue	Kafka broker not available
Category	Streaming
Service	Kafka
Severity	Critical
Status	Resolved
Description	Broker connection failing
Expected Resolution	Restart broker and check cluster health

Related Log Evidence

Log ID	Timestamp	Level	Message	Error Code
L011	5/1/2026 10:05	ERROR	Broker not available	KAFKA_BROKER
L027	5/1/2026 10:13	ERROR	Broker not available	KAFKA_BROKER
L047	5/1/2026 10:23	ERROR	Broker not available	KAFKA_BROKER

Procedure

1. Confirm broker unavailable errors in application logs and monitoring.
2. Identify impacted broker ID, host, namespace, and dependent applications.
3. Check Kafka service or pod status using systemctl or kubectl.
4. Review broker logs for connection refused, controller errors, disk full, or network failures.
5. Validate cluster health and look for offline partitions or leader value -1.
6. Restart the broker only after confirming process or pod health issue.
7. Validate Zookeeper or KRaft controller health depending on cluster mode.
8. Verify producer and consumer connectivity and monitor broker metrics for recurrence.

Common Commands

```
kafka-topics.sh --bootstrap-server <broker>:9092 --describe
kafka-consumer-groups.sh --bootstrap-server <broker>:9092 --describe --group <group>
kafka-broker-api-versions.sh --bootstrap-server <broker>:9092
kubectl get pods -n <namespace> | grep kafka
kubectl logs <kafka-pod> -n <namespace> --tail=100
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Broker unavailable	Kafka process stopped	Restart broker and validate health
Leader not available	Broker down or controller issue	Restore broker and check partition leaders
Connection refused	Port, DNS, or firewall issue	Validate endpoint and network path
Multiple brokers down	Infrastructure outage	Escalate to platform team

Best Practices

1. Maintain replication factor greater than one for production topics.
2. Monitor broker availability and offline partitions.
3. Document broker restart actions in the ticket.
4. Avoid restarting multiple brokers at the same time unless directed.

References

Apache Kafka Docs | Internal Logs CSV | Tickets CSV | Monitoring Dashboards | Kafka CLI Runbook